# SHM Worked Example 5.18

**Reference: Applied Mechanics: Reed's Vol. 2**

**Ex. 5.18**: A close-coiled helical spring has a stiffness of 1400 N/m. If a 10 kg mass is attached to the end of the spring, calculate the periodic time and the frequency of oscillation when the spring vibrates vertically.

## Solution

The equation for the natural frequency of a spring-mass system is given by:

$$\omega = \sqrt{\frac{k}{m}}$$

where:

- $k = 1400$ N/m (spring stiffness),
- $m = 10$ kg (mass).

Substituting the values:

$$\omega = \sqrt{\frac{1400}{10}} = \sqrt{140} = 11.8322 \text{ rad/s}$$

The periodic time $T$ is given by:

$$T = \frac{2\pi}{\omega}$$

Substituting $\omega = 11.8322$ rad/s:

$$T = \frac{2\pi}{11.8322} = 0.5310 \text{ s}$$

The frequency $f$ is given by:

$$f = \frac{1}{T}$$

Thus:

$$f = \frac{1}{0.5310} = 1.8831 \text{ Hz}$$

## Final Answers

- **Periodic Time**: $T = 0.5310$ s
- **Frequency**: $f = 1.8831$ Hz